

NUTRIENT AND PROXIMATE COMPOSITION OF PLANTAIN–ALMOND–OAT COMPOSITE FOODS FOR ELDERLY NUTRITION IN UMUDIKE, NIGERIA

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Abstract

*The increasing need for nutrient-dense functional foods for the elderly has stimulated interest in the utilization of locally available food resources. This study evaluated the nutrient and proximate composition of composite delicacies developed from plantain, almond, and oat for elderly consumers in Umudike, Ikwuano Local Government Area, Abia State, Nigeria. The purpose of the study was to develop innovative food products from plantain, oat, and almond nuts as a means of promoting diversification and enhancing their effective utilization. The study adopted research and development (R&D) design. research work was carried out in Umudike in South-East geopolitical zone Nigeria. The raw materials plantain (*Musa paradisiaca*), almond nuts (*Prunus amygdalus*), and oats (*Avena sativa*) were sourced from Ndoro Market in Umudike, Ikwuano LGA, Abia State. All samples were immediately transported in clean, sealed containers to the Home Economics Food Laboratory in Michael Okpara University of Agriculture, Umudike, where they were stored at ambient temperature (for plantains and almonds) and refrigerated (for oats) to preserve freshness before further processing and analysis. Plantains were washed in potable water, peeled, sliced into thin pieces (5–8 mm), and oven-dried at 60°C for 24 hours until crisp. Almond nuts were deshelled, sorted manually to remove broken seeds and foreign matter, and then roasted in a hot air oven at 150°C for 15 minutes to improve aroma and flavor. Oats were lightly steamed to reduce microbial load and enhance functionality, then dried at 50°C for 6 hours and milled to obtain oat flour. Each flour type was stored in airtight containers at room temperature until needed. These flours were then used in various ratios 60:20:20, 50:30:20, 40:30:30. Five products Almond Oat Plantain Bread (AOPB), Plantain Oat Cake (POB), Plantain Wrap (PW), Almond Oat Plantain Cake (AOPC), and Almond Oat Cake (POC) were formulated using standard food processing techniques. Micronutrient and proximate analyses were conducted using standard laboratory procedures, while sensory evaluation was performed using a 9-point hedonic scale. Data were analyzed using Analysis of Variance (ANOVA) at a 5% level of significance. The results showed significant differences ($p < 0.05$) in vitamin A ($F = 654.87$), vitamin B₁ ($F = 586.64$), and potassium ($F = 544.42$) among the formulations. Plantain recorded the highest vitamin A content, oats contained the highest vitamin B₁ concentration, while almonds supplied the highest potassium level. Moisture content ranged from 36.83–55.91%, protein 9.29–16.46%, crude fibre 0.34–1.41%, fat 4.69–16.46%, ash 0.90–1.57%, carbohydrate 22.80–45.79%, and energy 190.73–282.36 kcal/100 g. Sensory evaluation revealed significant differences ($p < 0.05$) among the products, with Almond Oat Plantain Bread recording the highest overall acceptability (37.00 ± 0.05). The study concludes that composite products from plantain, almond, and oat are nutritionally enriched, highly acceptable, and suitable as functional foods for elderly nutrition. Their production and consumption are therefore recommended.*

Keywords: Plantain, almond, oat, composite foods, micronutrients, proximate composition, sensory evaluation, elderly nutrition.

Introduction

Elderly population, particularly those in semi-rural communities like Umudike, Nigeria, are increasingly vulnerable to nutritional deficiencies due to a convergence of physiological, environmental, and socio-economic challenges. Ageing is often accompanied by decreased digestive efficiency,

appetite loss, reduced metabolic activity, and increased susceptibility to chronic illnesses, all of which contribute to poor dietary intake and malnutrition (Fetriyuna, Purwestri, Jati, and Setiawan, 2023). According to World Health Organization (WHO, 2021), the elderly are defined as persons above the age of 60 years with women comprising a majority of this population. The elderly population in the recent decade

especially in Africa and other developing countries appear to be increasing. Charlton Karen, Govender Vural, (2022) noted that the elderly are the gemstones of any society that are often ignored. Their care and wellbeing especially in rural communities depend largely on their children, relatives and sometimes government resources. Sanya (2021), have documented poor nutritional status among the aged. Similarly, previous studies such as Tucker (2022), have documented that the energy and nutrient intakes of the elderly were low compared to recommended dietary allowances.

According to Akinyele and Olatona (2022), many elderly individuals in Nigeria consume monotonous diets that fail to meet daily micronutrient requirements. Thus, profiling and determining the micronutrient contents of locally available and acceptable foods like plantain, almond, and oat is not only critical for dietary planning but also for designing targeted food interventions in communities like Umudike. Plantain is a starchy fruit widely consumed in Africa and plays a significant role in the diets of the elderly due to its digestibility and low allergenic potential. According to Onyeke, Onodugo, and Eze (2024), the micronutrient profile of unripe plantain includes potassium (approximately 499 mg/100g), magnesium (37 mg), iron (0.6 mg), calcium (3 mg), and moderate levels of vitamin C (18.4 mg). Almonds on the other hand are among the most nutrient-dense nuts globally and offer a rich profile of vitamins and minerals necessary for elderly health. Okpala, Ezeagbe, Agu, and Igwe (2024) reported that almonds contain approximately 264 mg of magnesium, 269 mg of calcium, 3.7 mg of iron, and 25.6 mg of vitamin E per 100 grams. They are also a good source of zinc (3.1 mg), folate, and B2 (riboflavin), all of which support immune function, cognitive integrity, and antioxidant defence. Oats are widely consumed for their cardiovascular benefits, but their micronutrient richness also merits attention. Alabi (2023) identified oats as an excellent source of manganese (4.9 mg/100g), phosphorus (410 mg), magnesium (138 mg), and iron (4.3 mg), as well as moderate amounts of B-vitamins including thiamine and pantothenic acid. These micronutrients are vital in energy metabolism, cognitive health, and red blood cell formation.

Micronutrient determination requires rigorous laboratory procedures to ensure accurate profiling. The Association of Official Analytical Chemists (AOAC, 2024) outlines standard procedures for analyzing vitamins and minerals in food samples. AAS is commonly used for elements like iron, zinc, calcium, and magnesium, while flame photometry is applied for sodium and potassium. Proximate composition analysis evaluates the levels of key macronutrients moisture, ash, fat, protein, fiber, and

carbohydrates and provides foundational data for assessing the nutritional adequacy of foods. The significance of these parameters in elderly nutrition is profound, as older adults require more digestible protein, lower saturated fat, and higher dietary fiber for optimal metabolic health (Özer, Özbül, & Özpala, 2022). For instance, combining plantain and oats with almond can result in composite meals with a balanced protein-energy ratio, improved fatty acid profile, and enhanced mineral content. According to Onyeke, Onodugo, and Eze (2024), oat-plantain blends demonstrated higher protein and fiber content compared to traditional cassava-based meals, underscoring the nutritional enhancement potential of these combinations.

Moisture content is a critical determinant of food shelf life, microbial stability, and texture. In elderly diets, foods with moderate to low moisture content are often preferred due to their storability and ability to reduce microbial spoilage. According to Onyeke, Onodugo, and Eze (2024), raw plantain has a moisture content of approximately 60–65%, which significantly reduces after drying or cooking. Almond nuts, being oil seeds, have a lower moisture content (around 5–6%) which contributes to their shelf stability (Okpala, Ezeagbe, Agu, & Igwe, 2024). According to Alabi (2023), oats contain 11–15% protein, with a favorable amino acid profile rich in lysine and methionine. Almonds offer even higher protein values, ranging between 19–21% depending on processing. Plantain, in contrast, contains lower protein (2.3–3.5%), but serves as a good carrier in composite blends. Protein determination is typically performed using the Kjeldahl method, which measures nitrogen content and converts it to protein by a factor of 6.25. Protein fortification using almond or oat flour in plantain-based meals helps meet the Recommended Dietary Allowance (RDA) for elderly individuals, which ranges between 0.8–1.2 g/kg body weight per day (Akinyele & Olatona, 2022).

Almonds are especially rich in monounsaturated fats, containing approximately 50–54% crude fat, which supports cardiovascular health and provides essential fatty acids like linoleic acid (Okpala, Ezeagbe, Agu, & Igwe, 2024). Oats contain around 6–9% fat, mostly polyunsaturated in nature. In contrast, plantain is very low in fat (~0.3–0.5%), making it ideal as a carbohydrate base in low-fat formulations. Plantain contains moderate amounts of crude fiber (1.5–2.2%), which increases upon ripening. Oats are an excellent source of both soluble and insoluble fiber, with a total crude fiber content of 10–12%, predominantly beta-glucans that have cholesterol-lowering properties (Alabi, 2023). Almonds provide 12–14% fiber, making them one of the

richest sources among nuts. Fiber determination is based on sequential digestion using dilute acid and alkali followed by incineration. According to Onyeke, Onodugo, and Eze (2024), plantain has an ash content of around 1.2–2.0%, almonds have 2.5–3.0%, and oats average 1.5–2.5%. Ash values vary depending on soil conditions and post-harvest processing. High ash content indicates better mineral availability especially essential ones like calcium, potassium, magnesium, and zinc.

Despite the rising population of the elderly in Nigeria, most food systems continue to cater for the general populace, with little regard for the specialized nutritional needs of aging individuals. These challenges are compounded in rural and semi-urban communities like Umudike, where economic constraints, low dietary diversity, and limited access to fortified foods further aggravate the risk of malnutrition.

Current dietary options for elderly individuals in Umudike are largely starchy, energy-dense meals made from cassava, yam, or rice, which often lack sufficient protein, vitamins, minerals, and essential fatty acids. While synthetic supplements are available, they are largely unaffordable and unsustainable for long-term use. At the same time, nutrient-rich local ingredients such as unripe plantain, almond seeds, and oat remain underutilized in food formulations tailored for the elderly. These crops are not only accessible and culturally familiar but also possess nutritional profiles that could be synergistically combined to create balanced meals. However, the absence of scientific data on their combined proximate and micronutrient composition, as well as sensory attributes in prepared delicacies, poses a barrier to their effective utilization. A lot of research works have been carried out by various individuals and organizations to explore partial or total substitute for plantain, oat, and almond nuts for bread, cake, and moi-moi products. Although plantain, oat, and almond nuts possess several advantages such as availability, accessibility, affordability, and high nutritional value, their potential remains underutilized. These food materials are primarily consumed in their raw or basic forms, with little to no value addition or secondary economic application. Therefore, this research seeks to develop innovative food products from plantain, oat, and almond nuts as a means of promoting diversification and enhancing their effective utilization. The findings of this study will enable Nigerians particularly, households in Umudike to rely more on the local food products to boost Nigeria economy. The findings of this research when made available will expand households; food choice and improve family's menu. It is believed that the

homemaker influences to a large extent the feeding pattern of the family members. This is for the fact that, they will begin to diversify the preparation of local foods to encourage its consumption.

Methodology

The study adopted the quasi-experimental and Research and Development (R&D) designs. Research and development (R&D) refers to innovation activities undertaken by individual organization or government in developing new services or products, or improving existing services or products. The research and development study usually involves some number of phases which are generally called R&D cycle (Hall, 2006). This study adopted Research and Development (R & D) of Gall, Gall and Borg (2010).

The research work was carried out in Umudike in South-East geopolitical zone Nigeria. Umudike community is made up of many households both of academics and non-academic staff. It has a longitude 420°E to 500°E and latitudes starting from 70,000N to 190,000N (Joanh, 2021). The raw materials plantain (*Musa paradisiaca*), almond nuts (*Prunus amygdalus*), and oats (*Avena sativa*) were sourced from Ndoro Market in Umudike, Ikwuano LGA, Abia State. Mature, unripe plantains were selected based on their firmness, green color, absence of bruises, and freedom from fungal or insect damage. Preference was given to freshly harvested bunches sold directly by smallholder farmers to ensure optimal freshness and nutrient retention. Almond nuts, though not cultivated extensively in Umudike, were sourced from specialty nut vendors and food ingredient distributors in Umuahia main market.

All samples were immediately transported in clean, sealed containers to the Home Economics Food Laboratory in Michael Okpara University of Agriculture, Umudike, where they were stored at ambient temperature (for plantains and almonds) and refrigerated (for oats) to preserve freshness before further processing and analysis.

Plantains were washed in potable water, peeled, sliced into thin pieces (5–8 mm), and oven-dried at 60°C for 24 hours until crisp. The dried slices were milled using a Hammer mill and sieved through a 0.5 mm mesh sieve to obtain fine plantain flour. Almond nuts were shelled, sorted manually to remove broken seeds and foreign matter, and then roasted in a hot air oven at 150°C for 15 minutes to improve aroma and flavor. After cooling, the nuts were ground into a fine paste using a stone grinder, and excess oil was pressed out using cheesecloth to yield a consistent flour-like texture. Oats

were lightly steamed to reduce microbial load and enhance functionality, then dried at 50°C for 6 hours and milled to obtain oat flour. This was done to preserve the integrity of β -glucan, the fiber component vital to cardiovascular health. Each flour type was stored in airtight containers at room temperature until needed. These flours were then used in various ratios (e.g., 60:20:20, 50:30:20, 40:30:30) for the production of (Plantain Bread, Plantain Oat Cake, Plantain Wrap, Almond Oat Plantain Cake, Almond Oat Cake aimed at enhancing nutritional density, functional value, and sensory acceptability for elderly consumers.

This methodical approach to sample collection ensures that all raw materials used in recipe development of Plantain Bread, Plantain Oat Cake, Plantain Wrap, Almond Oat Plantain Cake, Almond Oat Cake are consistent in quality, freshness, safety, and nutritional potential, thus minimizing variability in experimental outcomes and enhancing the reliability of the research (Afolabi, Olorode, & Adelekan, 2020; AOAC, 2019). The result of the nutrient composition analysis conducted was subjected to statistical analysis of variance (ANOVA) using SPSS Statistical tool at $p < 0.05$ and the mean was compared using Duncan's multiple range test.

RESULT

Table 1. Micronutrient in Plantain, Almond Nuts and Oat

Micronutrients	Source	Sum of Squares	Df	Mean Square	F-value	P-value
Vitamin A	Between Groups	25,114.22	2	12,557.11	654.87	<0.00
	Within groups	192.24	3	32.04	-	-
	Total	25,306.46	5	-	-	-
Vitamin B (Thiamine)	Between groups	0.953	2	0.476	586.64	<0.00
	Within groups	0.005	3	0.0081	-	-
	Total	0.9535	5	-	-	-
Potassium (K)	Between groups	136,232.41	2	68,116.21	544.42	<0.00
	Within groups	750.30	3	125.05	-	-
	Total	136,982.71	5	-	-	-

Values are $\pm$ SD values on the same column with different superscript are significantly different ($P < 0.05$)

The table 1 above present in the ANOVA table of the micronutrients found in Plantain, almond nut and oat. The result reveal significant variations ($p < 0.001$) in the concentrations of Vitamin A, Vitamin B1 (Thiamine), and Potassium (K) among the five food samples analyzed plantain (Almond Oat Plantain Bread, Plantain Oat Cake,

Plantain Wrap, Almond Oat Plantain Cake, Almond Oat Cake) made from blending plantain, almond nuts and oat. The large F-values recorded across all nutrients indicate strong between-group differences, confirming that each food source exhibits unique micronutrient characteristics.

Table 2 Proximate Composition of Different Delicacies produced from plantain, almond nuts and oats

Samples	Moisture Content (%)	Crude Protein (%)	Crude Fibre (%)	Fat (%)	Ash Content (%)	Carbohydrate (%)	Energy Value (Kcal)
PW	36.83 $\pm$ 0.00 ^e	9.29 $\pm$ 0.00 ^d	0.34 $\pm$ 0.00 ^c	4.69 $\pm$ 0.00 ^d	0.90 $\pm$ 0.00 ^d	22.80 $\pm$ 0.00 ^d	190.73 $\pm$ 0.00 ^d
AOPC	44.58 $\pm$ 0.00 ^d	10.18 $\pm$ 0.00 ^c	0.96 $\pm$ 0.00 ^c	6.21 $\pm$ 0.00 ^c	1.31 $\pm$ 0.00 ^c	25.48 $\pm$ 0.00 ^c	225.99 $\pm$ 0.00 ^c
POC	47.20 $\pm$ 0.00 ^c	10.75 $\pm$ 0.00 ^c	1.23 $\pm$ 0.00 ^b	6.83 $\pm$ 0.00 ^c	1.37 $\pm$ 0.00 ^c	26.05 $\pm$ 0.00 ^b	268.23 $\pm$ 0.00 ^b
AOPB	48.46 $\pm$ 0.00 ^b	15.66 $\pm$ 0.00 ^b	1.26 $\pm$ 0.00 ^b	11.51 $\pm$ 0.00 ^b	1.49 $\pm$ 0.00 ^b	26.94 $\pm$ 0.00 ^b	281.87 $\pm$ 0.00 ^a
POB	55.91 $\pm$ 0.00 ^a	16.46 $\pm$ 0.00 ^a	1.41 $\pm$ 0.00 ^a	16.46 $\pm$ 0.00 ^a	1.57 $\pm$ 0.00 ^a	45.79 $\pm$ 0.00 ^a	282.36 $\pm$ 0.00 ^a

Values are $\pm$ SD values on the same column with different superscript are significantly different ($P < 0.05$)

Result in table 4.2 above shows the proximate composition of the delicacies (Almond Oat Plantain Bread, Plantain Oat Cake, Plantain Wrap, Almond Oat Plantain Cake, Almond Oat Cake) made from blending plantain, almond nuts and oat. The compositional data in the table demonstrate a progressive nutritional enhancement from PW to POB, with POB consistently recording the highest levels of protein, fat, fibre, carbohydrate, and energy value. This indicates that POB may offer greater nutritional and calorific benefits, albeit with higher moisture content that could affect its storage stability. The differences across samples suggest the influence of processing variations or ingredient formulations on proximate composition and nutritional quality.

Key: PW = Plantain Wrap, AOPC = Almond Oat Plantain Bread, POC = Plantain Oat Cake, AOPB = Almond Oat Plantain Bread, POB = Plantain Oat Bread

Discussion

This study sought to determine the micronutrients and proximate composition of different delicacies (Almond Oat Plantain Bread, Plantain Oat Cake, Plantain Wrap, Almond Oat Plantain Cake, Almond Oat Cake) made from blending plantain, almond nuts and oat. The analysis revealed that there are different micronutrients found in Plantain, almond nuts and oats, however, three were found to be more relevant as it pertains to the elderly in Umudike, Ikwuano Local Government Area of Abia State. Which included vitamin A, vitamin B1 and potassium. These micronutrients contained essential nutrients capable of supporting vision and immune function, aiding carbohydrate metabolism and regulating fluid balance and blood pressure

The findings demonstrate that composite foods produced from plantain, almond, and oat possess considerable nutritional potential for elderly nutrition. Significant variations in vitamin A, vitamin B₁, and potassium among the formulations reflect the complementary nutrient contributions of the three ingredients. The high vitamin A content observed in plantain-based products corroborates the report of Onyeke, Onodugo, and Eze (2024) that plantain is an important source of essential micronutrients, while the high thiamine content of oat-containing products supports the findings of Alabi (2023), who identified oats as a rich source of B-vitamins required for energy metabolism and neurological function. Similarly, the higher potassium content associated with almond-enriched formulations agrees with Okpala, Ezeagbe, Agu, and Igwe (2024), who reported that almonds are nutrient-dense foods containing substantial amounts of

essential minerals required for cardiovascular and metabolic health.

The proximate composition further showed that blending plantain, almond, and oat improved the nutritional quality of the products through increased protein, fibre, fat, ash, carbohydrate, and energy contents. These findings support the assertion of Özer, Özbuldu, and Özpapas (2022) that composite foods combining cereals, fruits, and nuts provide balanced nutrient profiles suitable for elderly nutrition. The relatively higher protein content of almond- and oat-based formulations also agrees with Alabi (2023) and Akinyele and Olatona (2022), who reported that oats and almonds enhance dietary protein intake, thereby contributing to the recommended protein requirements of older adults. Likewise, the increased fibre and ash contents are consistent with the reports of Onyeke et al. (2024) and Alabi (2023), who emphasized the role of these ingredients in improving digestive health and mineral availability among the elderly.

Conclusion

This study demonstrated that composite products developed from plantain, almond, and oat are nutritionally enriched and acceptable for elderly consumption. The formulations differed significantly in their micronutrient and proximate compositions, with plantain contributing higher vitamin A, oats providing vitamin B₁, and almonds supplying substantial potassium. Among the products evaluated, Almond Oat Plantain Bread (AOPB) exhibited the most desirable nutritional profile and the highest overall sensory acceptability. The findings indicate that blending these locally available food materials can produce functional foods capable of improving nutrient intake while maintaining desirable sensory qualities for older adults.

Recommendations

Based on the findings of the study, the following recommendations are made:

1. Almond Oat Plantain Bread (AOPB) should be promoted for household and commercial production because of its superior nutritional quality and high consumer acceptability.
2. Nutritionists, healthcare providers, and relevant government agencies should encourage the inclusion of plantain–almond–oat composite foods in the diets of elderly individuals to improve micronutrient intake and overall nutritional status.
3. Further studies should investigate the shelf life, storage stability, and long-term health benefits

of these composite products to facilitate their large-scale production and commercialization.

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